Dr. ABDUL RAZAK

10+ years of experiences in AI and Machine Learning Research

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EXPERIENCE

Senior Data Scientist **June 2023 - Present**

Extreme Networks Salem, United States

Following are some of the areas that I am currently focusing on to build AI/ML/DL based approaches for wireless networking:

- Designed and deployed GenAI-based diagnostic models to analyze complex wireless client behavior, roaming issues, and device performance anomalies in enterprise networks.
- Applied multi-modal AI to correlate network telemetry, device health indicators, and client performance data for accurate root-cause identification.
- Implemented network-aware GenAI inference pipelines that integrate statistical models, and LLM reasoning to evaluate anomalies over time and reduce false alerts.
- Applied GenAI techniques to evaluate client experience metrics such as connectivity reliability, roaming success, throughput stability, and latency trends.
- Improved network troubleshooting efficiency by developing GenAI-powered tools that reduce investigation time for wireless issues from hours to seconds.
- Collaborated with cross-functional engineering and product teams to embed GenAI troubleshooting capabilities into enterprise networking platforms.
- Design autonomous agent workflows that combine GenAI, traditional ML, and domain knowledge graphs to support continuous network monitoring and automated remediation planning.
- Enhanced customer experience by enabling AI-driven identification of performance bottlenecks and configuration inconsistencies across thousands of APs and switches.
- Develop preference elicitation and handling techniques to model user preferences to provide the users with personalized wireless networking solutions.
- Participate in optimizing the existing algorithms in the products to improve accuracy and efficiency.
- Apply multi-objective optimization techniques to solve wireless networking issues.
- Work with internal experts to build innovative AI and machine learning automated intelligent systems to measure the network performance, detect anomalies, resolve client or wireless/wired devices performance issues to improve overall client experience.
- Participate in building the NextGen real time AI/ML pipelines
- Investigate and propose emerging/cutting edge, open source AI/ML algorithms and platforms to improve existing projects and ensure that data science projects align with organisational goals.
- Apply exact and approximate AI/ML/DL algorithms/techniques to build data driven solutions for complex wireless and wired networking problems.
- Probabilistic Programming: PyMC3
- Languages: Python, Pyspark, PyFlink and SQL
- ML platform: Azure, AWS Athena and SageMaker

Senior Data Scientist **Jan 2020 - May 2023**

Extreme Networks Cork, Ireland

Following are some of the areas that I worked in AI/ML/DL based approaches for wireless networking:

- Work with internal experts to build innovative AI and machine learning automated intelligent systems to measure the network performance, detect anomalies, resolve client or wireless/wired devices performance issues to improve overall client experience.
- Participate in building the NextGen real time AI/ML pipelines
- Work with engineers and product managers to turn data science prototypes into robust and reliable solutions.

- Investigate and propose emerging/cutting edge, open source AI/ML algorithms and platforms to improve existing projects and ensure that data science projects align with organisational goals.
- Apply exact and approximate AI/ML/DL algorithms/techniques to build data driven solutions for complex wireless and wired networking problems.
- Maintain and tune the algorithms/models for the machine learning pipeline.
- Probabilistic Programming: PyMC3
- Languages: Python, Pyspark, PyFlink and SQL
- ML platform: AWS Athena and SageMaker

Data Scientist **Jun 2017 - Jan 2020**

Johnson Controls Cork, Ireland

ML/AI techniques were implemented for building technologies such as intrusion alarm and access control; retail analytics such as digital signage:

- Exact and approximate AI algorithms
- Supervised and unsupervised machine learning
- Probabilistic Programming: BLOG (Bayesian Logic)
- Languages: Python, Pyspark and SQL
- ML platform: Azure Databricks

Post Doctoral Researcher (AI/ML) **May 2014 - May 2017**

Insight Centre for Data Analytics Cork, Ireland

My research work was a part of United Technology Research Center's (link) Machine Learning research project. In addition, I worked partly in collaboration with IBM Research Labs in Dublin.

- Probabilistic Programming Languages (PPLs) for developing probabilistic graphical models
- Clustering Algorithms
- Decision Making Problems Under Uncertainty (Influence Diagrams)
- Other Machine Learning Techniques such as regression analysis
- Languages: Python and PPL (BLOG and Swift)

PhD Researcher (Artificial Intelligence) **Nov 2010 - Apr 2014**

University College Cork (UCC) Cork, Ireland

My research work was part of one of the three main streams of IBM's Risk Management Colloboratory research project (link). I mainly worked on building and developing the techniques for handling user preferences. Apply these preference models to solve very complex multi-objective decision making problems, in situations where everything is known or unknown prior to making the decisions. My roles in this project include:

- Investigate existing techniques/algorithms for extracting user preference information (in multiple objective settings)
- Enhance the existing exact and approximate algorithms with the preference information
- Conduct experiments and compare the results against existing techniques/algorithms
- Languages: C++ and Python
- C++ libraries: STL, Boost, Eigen, Linear Programming Solvers (ILOG CPLEX and LPSOLVE)

Lecturer (Mathematics) **May 2005 - Jun 2010**

Osmania University Hyderabad, India

I am passionate about teaching mathematics, during this time I taught the following topics in mathematics for undergraduate students:

- Abstract and Linear Algebra
- Probability theory and statistics
- Differential Calculus
- Numerical Analysis
- Real Analysis
- Differential Vector Calculus
- Co-ordinate Geometry

PATENTS (USA)

Google Patents List

- Interactive method to reduce the amount of tradeoff information required from decision maker in multi-attribute decision making under uncertainty, Abdul Razak, Lea A Deleris, Radu Marinescu, Peter Nicholas Wilson, 2019.
- Virtual Maintenance Manager, Abdul Razak, Gopi Subramanian, Conor Joseph Donovan, 2019.
- Automated Alarm Panel Classification Using Pareto Optimization, Abdul Razak, Michael Stewart, Gopi Subramanian, 2020.
- Systems and Methods for Managing a Security System, Marcin Dziduch, Conor Joseph, Donovan, Abdul Razak, 2020.
- Rules-based method of identifying misuse of emergency fire exists using data generated by a security alarm system, Marcin Dziduch, Conor Joseph, Donovan, Abdul Razak, 2020.
- Building security analysis system with site-independent signature generation for predictive security analysis, Marcin Dziduch, Conor Joseph, Donovan, Abdul Razak, 2020.
- Systems and methods for managing alarm data of multiple locations, Abdul Razak, 2021.
- Building security system with alarm lists and alarm rule editing, Maebh Costello, Federico Fala, Jan R. Holliday, Nicolae Bogdan Pavel, Abdul Razak, 2021
- Root-cause analysis of event occurrences, Donagh Horgan, Abdul Razak, Giacomo Bernardi, 2021

SCIENTIFIC PUBLICATIONS IN INTERNATIONAL CONFERENCES

Google Scholar Profile (Link)

- Multi-objective Influence Diagrams with Possibly Optimal Policies, Radu Marinescu, Abdul Razak, Nic Wilson in Proceedings of the Thirty-first AAAI Conference on Artificial Intelligence 2017 (AAAI 2017).
- The Comparison of Multi-Objective Preference Inference Based on Lexicographic and Weighted Average models, Anne-Marie, Abdul Razak and Nic Wilson in Proceedings of the Twenty Seventh IEEE International Conference on Tools with Artificial Intelligence 2015 (ICTAI 2015).
- Multi-Objective Constraint Optimisation with Lexicographic Preference Models, Anne-Marie, Abdul Razak and Nic Wilson workshop on Algorithmic issues for Inference in Graphical Models 2015 (AIGM 2015).

- Computing Possibly Optimal Solutions for Multi- Objective Constraint Optimisation with Tradeoffs, Nic Wilson, Abdul Razak and Radu Marinescu in Proceedings of the Twenty Fourth International Joint Conference on Artificial Intelligence 2015 (IJCAI 2015).
- Multi-objective constraint optimisation with tradeoffs, Radu Marinescu, Abdul Razak and Nic Wilson in Proceedings of the Nineteenth International Conference on Principles and Practice of Constraint Programming 2013 (CP 2013).
- Multi-objective influence diagrams with tradeoffs, Radu Marinescu, Abdul Razak and Nic Wilson in Proceedings of the Twenty-eighth Conference on Uncertainty in Artificial Intelligence 2012 (UAI 2012).

SCIENTIFIC REVIEWING ACTIVITY

- **Reviewer**, *Computers & Electrical Engineering* (Elsevier); completed 10+ journal manuscript reviews in the areas of AI, wireless networking, and intelligent systems.
- **Reviewer**, IEEE International Conference on Communications (ICC); evaluated AI-based research submissions for the 2025 conference technical program.

EDUCATION

PhD in Computer Science (AI)	**Nov 2010 - May 2014**
University College Cork	Cork, Ireland
MSc in Mathematics with Computer Science	**Jun 2003 - May 2005**
Osmania University	Hyderabad, India
BSc in Mathematics, Statistics and Computer Science	**Jun 2000 - Mar 2003**
Osmania University	Hyderabad, India

TRAINING

Attended various international summer schools for learning techniques and algorithmic aspects in the field of Artificial Intelligence and Machine Learning as part of my PhD program.

AWARDS AND ACHIEVEMENTS

- Received Science Foundation Ireland (SFI) Post Doctoral Fellowship award for the years 2014-2017 at Insight Centre for Data Analytics, Computer Science Department, University College Cork, Cork, Ireland.
- Received IRCSET IBM Enterprise Partnership Scheme (EPS) Scholarship award for the years 2010-2013 that supported me to complete my PhD degree at Cork Constraint Computation Centre, Computer Science Department, University College Cork, Cork, Ireland.
- Received Osmania University merit scholarship to pursue two years of Masters degree between 2003 - 2005.
- Secured 40th Rank in a mathematics entrance test conducted by Osmania University in the year 2003 for the admission into the master degree in Mathematics with Computer Science. Approximately 30 thousand students appeared for this test, where only 120 seats were available in the main campus of Osmania University.

TALKS

- Presentations on preference handling techniques in decision making problems with certainty and uncertainty at Sapienza University (Italy), University College Cork (Ireland), National University of Ireland Galway (Ireland) and IBM research labs in Dublin.
- A number of talks during my AI/ML project work with United Technology Research Centre in Ireland.
- A number of oral and poster presentations at various international conferences and workshops.

PERSONAL DETAILS

Nationality	Irish (since May, 2021)
Date of Birth	25th June 1980
Birth Nationality	Indian
Martial Status	Married
Languages	English (read-write-spoken), Hindi (read-write-spoken),
	Telugu (read-write-spoken) Urdu (read-spoken)